TITANIC SURVIVAL PREDICTION USING MACHINE LEARNING

Github Link: https://github.com/Nutana09/Titanic-Survival-Prediction

Introduction

The Titanic disaster is one of the most well-known tragedies in history and has become a popular dataset for introducing machine learning concepts. The dataset contains demographic and travel-related information of passengers aboard the Titanic, along with their survival status.

Predicting survival based on available features is a classic classification problem in machine learning. It helps in understanding data preprocessing, feature engineering, and classification algorithms. Such predictive models are widely applicable in real-world scenarios such as risk assessment, medical prognosis, customer churn prediction, and decision-making systems.

Problem Statement

The objective of this project is to build a machine learning model that predicts whether a passenger survived the Titanic disaster based on demographic and ticket-related information.

- **Input Features:** Age, Sex, Passenger Class (Pclass), Fare, Family Size
- **Target Variable:** Survived (1 = Survived, 0 = Did Not Survive)

The model uses the **Weighted K-Nearest Neighbors (WKNN)** algorithm to perform classification.

Dataset Description

The dataset used in this project is the Titanic Survival Prediction dataset, commonly available on Kaggle. It contains information about 891 passengers.

Key Features:

- **Pclass:** Passenger class (1 = First, 2 = Second, 3 = Third)
- **Sex:** Gender of the passenger
- **Age:** Age in years
- **Fare:** Ticket fare paid
- **SibSp:** Number of siblings/spouses aboard
- **Parch:** Number of parents/children aboard

Feature Engineering:

A new feature **Family Size** is created:

$$FamilySize = SibSp + Parch + 1$$

This helps capture family relationships, which may influence survival chances.

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

Figure 1: Dataset Sample output

```
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Embarked       0
FamilySize     0
dtype: int64
```

Figure 2: Missing values output

Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand patterns and relationships in the dataset.

Steps Performed:

- Checked for missing values
- Visualized survival distribution
- Analyzed survival rate based on gender and class

Key Observations:

- Female passengers had a significantly higher survival rate than males
- Passengers in higher classes (1st class) had better survival chances
- Younger passengers (children) were more likely to survive
- Missing values were present in the Age column

These insights helped guide feature selection and preprocessing.

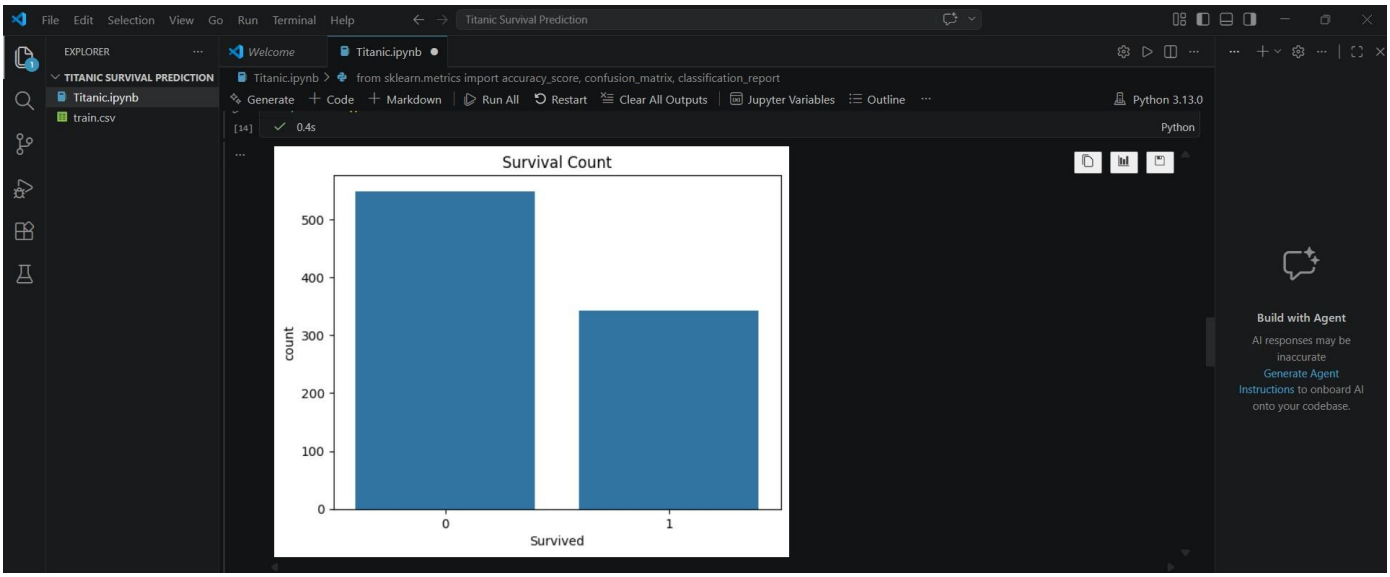

Figure 3: Survival Count graph output

Figure 4: Survival by Gender output

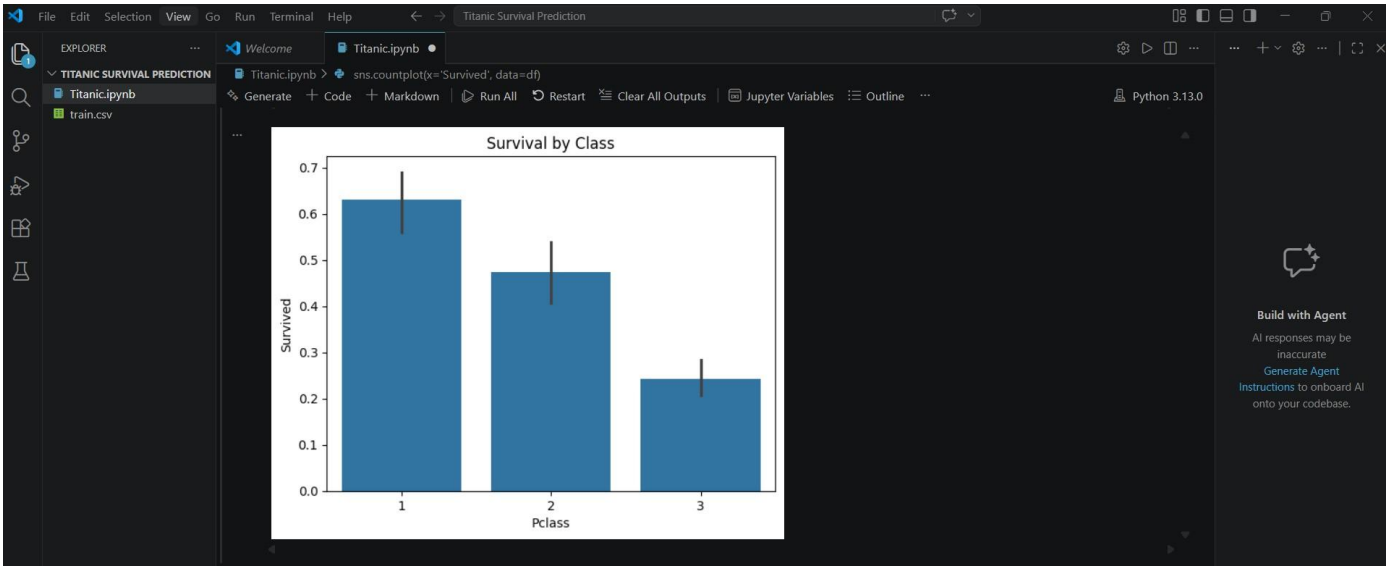

Figure 5: Survival by Class output

Methodology (Implementation)

Data Preparation

- Missing values in Age and Fare were filled using median values
- A new feature (FamilySize) was created

Feature Transformation

- Categorical variable "Sex" was encoded into numerical values:
  - Male → 0
  - Female → 1

Feature Selection

The following features were selected:

- Age
- Sex
- Pclass
- Fare
- FamilySize

Data Splitting

The dataset was split into training and testing sets using an 80-20 ratio.

Feature Scaling

Since KNN is distance-based, feature scaling was applied using StandardScaler to normalize the data.

Model Selection: WKNN

The **Weighted K-Nearest Neighbors (WKNN)** algorithm was used. Unlike standard KNN, WKNN assigns higher importance to closer neighbors using distance-based weighting.

Model Training

The model was trained using:

- Number of neighbors (k) = 5
- Weight function = distance

Model Evaluation

The model was evaluated using:

- Accuracy Score
- Confusion Matrix
- Classification Report (Precision, Recall, F1-score)

Results and Evaluation

The WKNN model achieved an accuracy of approximately **75% to 80%** on the test dataset.

Confusion Matrix Analysis

- The model correctly predicted a large number of non-survivors
- Some survivors were misclassified, indicating moderate recall for the positive class

Performance Metrics • Accuracy: ~78%

- **Precision:** Good for non-survivors
- **Recall:** Moderate for survivors

Overall, the model performed reasonably well for a simple classification algorithm.

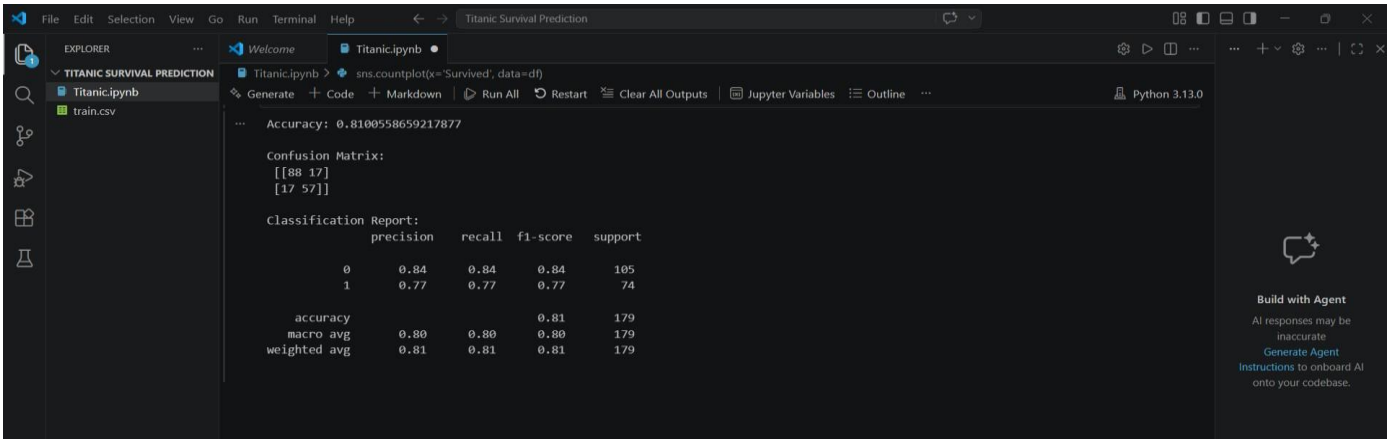

Figure 6: Accuracy, Confusion Matrix, Classification Report output

Discussion

Strengths:

- Simple and easy to implement
- No assumptions about data distribution
- WKNN improves performance over standard KNN by weighting neighbors
- **Limitations:**
- Sensitive to feature scaling
- Performance depends on choice of k
- Computationally expensive for large datasets
- Does not handle missing values directly
- **Improvements:**
- Hyperparameter tuning for optimal k value
- Use of advanced models like Random Forest or Logistic Regression
- Better feature engineering

Applications

The techniques used in this project can be applied in various real-world scenarios:

- Healthcare (patient survival prediction)
- Banking (fraud detection)
- Marketing (customer churn prediction)
- Insurance (risk assessment)

Conclusion

This project successfully implemented a classification model using the Weighted K-Nearest Neighbors algorithm to predict Titanic passenger survival. The results demonstrate that proper data preprocessing, feature engineering, and model selection significantly impact performance. Although WKNN is a simple algorithm, it provides decent accuracy and serves as a strong baseline model. The project highlights the importance of understanding data and applying appropriate machine learning techniques to solve realworld problems.

References

1.Kaggle Titanic Dataset

2.Scikit-learn Documentation

3.Introduction to Machine Learning – Textbooks

4.Online tutorials and research articles on KNN